

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a139de873639770ba.jsonl`  
- SHA-256: `487ddf2141b9afa3f56a8f88b84a30f5c963539cd2d926b108a98e3e93a3493f`  
- Source modified: `2026-05-31T04:22:03+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.049Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Pixtral-12B scores 90.7 ANLS on DocVQA and 81.8 (CoT) on ChartQA, indicating strong document-understanding/chart-reading capability relevant to parsing statement tables."

****Source:**** <https://huggingface.co/mistralai/Pixtral-12B-2409> (primary)

****Supporting quote:**** "DocVQA (ANLS): 90.7 ... ChartQA (CoT): 81.8"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:09.899Z)

I'll verify this claim about Pixtral-12B's benchmark scores. Let me start by checking the primary source and looking for the actual model card content.

3. assistant (2026-05-31T04:21:13.316Z)

I have the tools loaded. Now let me fetch the primary source to verify the exact benchmark numbers claimed.

4. user (2026-05-31T04:21:19.035Z)

Benchmark Scores for DocVQA and ChartQA

DocVQA (ANLS)

****Score: 90.7**** for Pixtral-12B-2409

This appears in the ****Multimodal Benchmarks**** table, where Pixtral-12B-2409 is compared against:

- Qwen2 7B VL: 94.5
- LLaVA-OV 7B: 90.5
- Phi-3 Vision: 84.9
- Phi-3.5 Vision: 74.4

ChartQA (CoT)

****Score: 81.8**** for Pixtral-12B-2409

This also appears in the ****Multimodal Benchmarks**** table. ****Yes**, ChartQA uses CoT (Chain-of-Thought) as indicated by the `_(CoT)_` notation.

****Comparison models:****

- Qwen2 7B VL: 38.6
- LLaVA-OV 7B: 67.1
- Phi-3 Vision: 72.0
- Phi-3.5 Vision: 67.7

Pixtral-12B-2409 achieves the highest score on ChartQA among these open-source models, significantly outperforming Qwen2 7B VL and leading over most other comparable models.

5. user (2026-05-31T04:21:20.336Z)

Web search results for query: "Pixtral-12B DocVQA ANLS 90.7 ChartQA 81.8 benchmark"

Links: [{"title":"Pixtral 12B","url":"https://arxiv.org/html/2410.07073v2"}, {"title":"Pixtral 12B 2409 · Models · Dataloop","url":"https://dataloop.ai/library/model/mistralai_pixtral-12b-2409/"}, {"title":"Papers Explained 219: Pixtral. Pixtral is a 12B parameter natively? | by Ritvik Rastogi | Medium","url":"https://ritvik19.medium.com/papers-explained-219-pixtral-a714f94e59ac"}, {"title":"Use Pixtral 12B for multimodal AI queries in Python with watsonx | IBM","url":"https://www.ibm.com/think/tutorials/multimodal-ai-pixtral-12b-on-watsonx-ai"}, {"title":"--- library_name: vllm language: - en - fr - de - es - it - pt - ru - zh - ja","url":"https://huggingface.co/mistralai/Pixtral-12B-2409/raw/main/README.md"}, {"title":"Pixtral Large: Mistral AI's Groundbreaking Leap in Large Language Models | Encord","url":"https://encord.com/blog/pixtral-large-explained/"}, {"title":"Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)","url":"https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title":"Pixtral Large: Specs, Benchmarks & API Access Guide (2026)","url":"https://ucstrategies.com/news/pixtral-large-specs-benchmarks-api-access-guide-2026/"}, {"title":"Pixtral 12B ? Another great multimodal model | UnfoldAI","url":"https://unfoldai.com/pixtral-12b/"}, {"title":"Pixtral 12B","url":"https://arxiv.org/pdf/2410.07073"}]

Based on the search results, Pixtral-12B-2409 achieves DocVQA (ANLS) score of 90.7 and ChartQA (CoT) score of 81.8.

Pixtral is a 12B parameter natively multimodal vision-language model based on Mistral Nemo. For certain benchmarks, including DocVQA (ANLS) and VQAV2 (VQA Match), the model outperforms OpenAI's GPT-4o and Claude-3.5 Sonnet.

The benchmark scores you mentioned are confirmed in the official documentation and multiple sources discussing Pixtral's multimodal capabilities.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:23.248Z)

The numbers are confirmed by both the primary source (HuggingFace model card) and corroborating sources. Let me verify the exact wording on the raw README and check the arXiv paper to confirm

these are the official figures, and to scrutinize whether there's any qualification or dispute.

7. user (2026-05-31T04:21:29.476Z)

DocVQA and ChartQA Scores for Pixtral-12B-2409

Exact Lines from the Document

```
**DocVQA:**
...
| **DocVQA** *(ANLS)*          | 90.7          | **94.5**      | 90.5          | 84.9          | 74.4
|
...

**ChartQA:**
...
| **ChartQA** *(CoT)*          | **81.8**      | 38.6          | 67.1          | 72.0          | 67.7
|
...
```

Exact Numbers for Pixtral-12B-2409

- **DocVQA (ANLS):** 90.7
- **ChartQA (CoT):** 81.8

CoT Labeling

Yes, **ChartQA** is labeled as **CoT** (Chain of Thought), as shown in the table header:
``**ChartQA** *(CoT)*``

DocVQA uses the ***(ANLS)*** metric instead.

Official/Self-Reported Status

These scores are **self-reported by Mistral**. The document states at the beginning of the benchmarks section:

> "The performance of Pixtral-12B-2409 compared to multimodal models. **All models were re-evaluated and benchmarked through the same evaluation pipeline.**"

This indicates Mistral conducted their own evaluation rather than citing third-party results.

8. user (2026-05-31T04:21:30.994Z)

Pixtral 12B Performance Metrics

Based on Table 2 in the paper, Pixtral 12B achieves:

- **DocVQA (ANLS):** "90.7"
- **ChartQA (CoT):** "81.8"

Important Caveats

The paper emphasizes that these results come with significant evaluation protocol considerations. The authors note that "evaluation protocols for multimodal language models is not standardized, and that small changes in the setup can dramatically change the performance of some models."

Specifically, they found that many models underperformed their reported figures under the common evaluation protocol used here. The paper provides "Explicit" prompts that "explicitly specify the format required by the reference answer," which differs from default benchmark prompts. Some

models required flexible parsing metrics to match their reported numbers, though Pixtral 12B maintained strong performance without such adjustments.

9. assistant (2026-05-31T04:21:33.837Z)

I now have strong confirmation from multiple sources. Let me do one more skeptical check: search for any independent benchmark or critique that disputes Pixtral's document/table understanding, and specifically whether these scores translate to financial table parsing reliability.

10. user (2026-05-31T04:21:41.538Z)

Web search results for query: "Pixtral-12B document table extraction accuracy independent benchmark OCR critique"

Links: [{"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "Pixtral 12B or Llama-3.2-11B-Vision for OCR tasks involving ID photos or PDF documents | watsonx.ai", "url": "https://community.ibm.com/community/user/discussion/pixtral-12b-or-llama-32-11b-vision-for-ocr-tasks-involving-id-photos-or-pdf-documents"}, {"title": "A Guide to Invoice Parsing and Analysis Using Pixtral-12B Model for OCR and RAG - Superteams.ai", "url": "https://www.superteams.ai/blog/a-guide-to-invoice-parsing-and-analysis-using-pixtral-12b-model-for-ocr-and-rag"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/pdf/2410.07073"}, {"title": "Beyond OCR: How Multimodal AI is Changing the Game of Image Understanding | by Daniel Alexander Muro | Capgemini Invent Lab | Medium", "url": "https://medium.com/capgemini-invent-lab/from-ocr-to-multimodal-a-new-era-in-image-to-text-technology-8d45d7559f01"}, {"title": "[Deprecated] Pixtral 12B | Mistral AI", "url": "https://mistral.ai/news/pixtral-12b/"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}]

Based on the search results, here's what I found regarding Pixtral-12B's document table extraction accuracy and OCR performance:

Benchmark Performance

The 83.7% ChartQA score demonstrates strong OCR and structured data extraction for a 12B model. It can parse complex charts, extract table data from screenshots, and reason about visual trends with acceptable accuracy.

Independent Benchmark Comparison

In an independent evaluation comparing multiple models on information extraction tasks, Pixtral 12b performs slightly lower across both recall and precision metrics compared to Pixtral Large and Gemini 1.5 Flash. More specifically, the OCR model falls significantly behind, with a recall of just 26% and a precision of 61%, suggesting multimodal models like Pixtral outperform traditional OCR approaches.

Limitations

There are notable limitations to consider: OCR accuracy drops significantly on non-Latin scripts like Chinese, Arabic, or Cyrillic. Additionally, inference is 30-50% slower than text-only models at the same parameter count, with Pixtral running at 40 tokens per second on an RTX 4090 with Q5 quantization.

The model appears well-suited for business document processing tasks where it can extract structured data, though it performs better on English-language documents compared to multilingual content.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-05-31T04:21:52.382Z)

I now have sufficient evidence to evaluate the claim carefully.

Assessment:

The claim has two parts: (1) the benchmark numbers, and (2) the inference that these scores indicate "strong document-understanding/chart-reading capability relevant to parsing statement tables."

Part 1 ? Numbers are accurately quoted. The primary source (HuggingFace model card / raw README) and the arXiv paper (Table 2) both confirm exactly: DocVQA (ANLS) = 90.7 and ChartQA (CoT) = 81.8. The quote is verbatim. The claim does not overstate or misattribute the numbers, and correctly flags ChartQA as CoT.

Part 2 ? The interpretive leap is reasonable but worth noting. DocVQA and ChartQA are legitimately the standard benchmarks for document/chart understanding, so the inference "relevant to parsing statement tables" is sound at a directional level. The claim is appropriately hedged ("indicating... capability relevant to") rather than asserting Pixtral will reliably parse bank statements.

Skeptical caveats (qualify but do not refute):

- These are self-reported by Mistral, re-evaluated through Mistral's own pipeline. The paper itself acknowledges MM eval protocols are non-standardized and "small changes in setup can dramatically change performance." So absolute comparability is soft.
- DocVQA/ChartQA performance does NOT directly equal dense multi-column financial-table transcription accuracy (the research question's actual target). An independent eval found Pixtral-12B "slightly lower across both recall and precision" vs Pixtral Large and Gemini 1.5 Flash on extraction. But the claim never asserts table-parsing accuracy ? only "capability relevant to."

The claim is well-supported, current (model from late 2024, still relevant 2025-26), and sourced to a primary model card corroborated by the arXiv paper. The numbers are exact and the interpretation is appropriately hedged. This survives skeptical scrutiny.

12. user (2026-05-31T04:21:59.036Z)

Structured output provided successfully

13. assistant (2026-05-31T04:22:03.356Z)

Verification complete. The claim is NOT refuted (refuted=false, high confidence).

The benchmark numbers (DocVQA ANLS 90.7, ChartQA CoT 81.8) are quoted verbatim and confirmed by both the primary HuggingFace model card and the arXiv paper Table 2. ChartQA is correctly flagged as CoT. The interpretive claim is appropriately hedged ("indicating... capability relevant to") rather than overstating that Pixtral will reliably parse bank statements.

Caveats logged but non-refuting: scores are Mistral-self-reported via their own eval pipeline (the paper itself warns MM eval is non-standardized), and DocVQA/ChartQA scores do not directly equal dense financial-table transcription accuracy ? but the claim never asserts the latter.